

Saksham Garg

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PROFILE

- Quantitative research-focused engineer with experience in systematic alpha research, market microstructure, statistical modeling, and machine learning. Built research systems for cost-aware signal evaluation across 15 ETFs and 2,422 trading days and limit-order-book analysis using 400K+ NASDAQ market events. Strong foundations in probability, statistics, time-series analysis, C++, and Python, with experience in walk-forward validation, transaction-cost modeling, and out-of-sample testing.

PROJECTS

Systematic Alpha Research Lab – Cost-Aware Quantitative Signal Research [GitHub]

Python, Pandas, NumPy, SciPy, yfinance, Next.js, TypeScript

- Achieved 0.850 OOS Sharpe and 1.184 Sortino for cross-sectional momentum over 2,422 trading days (2017–2026), with t-stat = 2.64, $p = 0.008$, and 9.15% CAGR across 15 ETFs using walk-forward validation.
- Validated a multi-signal ensemble at 0.643 OOS Sharpe, delivering 6.88% CAGR and -16.37% maximum drawdown, while accounting for transaction costs and execution lag.
- Rejected short-term reversal after OOS Sharpe fell to 0.468 with 18.4× annualized turnover, using regime, crisis, and parameter-robustness testing to identify fragile signals.
- Reduced look-ahead and trading-cost bias using a 1-day execution lag, monthly rebalancing, 10 bps transaction costs + 5 bps slippage, cash fallback, and turnover tracking.

Real-Time Order Book Imbalance Predictor – High-Frequency Quant ML Engine [GitHub]

Python, Pandas, NumPy, SciPy, CatBoost, XGBoost, scikit-learn, Streamlit, WebSockets, SQLite, Parquet, SHAP

- Engineered 30+ high-frequency microstructure features across 20 L2 order-book depth levels, including multi-level Order Book Imbalance (OBI), micro-price deviation, exponential volume decay, and 100ms/500ms rolling volatility.
- Architected a low-latency streaming pipeline consuming Binance L2 WebSockets with async SQLite/Parquet snapshot persistence, processing microsecond-stamped data at 100ms update frequencies.
- Benchmarked CatBoost, XGBoost, and Logistic Regression for 1s/3s forward directional price prediction, achieving 0.0063 Log Loss with sub-50ms inference latency.
- Implemented SHAP model explainability and an interactive Streamlit dashboard to monitor real-time book imbalance, dynamic volume decay curves, feature importance, and model calibration.

EXPERIENCE

Affy Pharma Pvt. Ltd.

May 2026 – Jul. 2026

AI/ML Intern

Noida, India

- Reduced manual data-preparation effort by 35% by building automated preprocessing and cleaning pipelines for manufacturing and quality-control records.
- Achieved up to 91% validation accuracy using Random Forest, XGBoost, and Logistic Regression for predictive analytics and model comparison.
- Improved evaluation reliability by standardizing preprocessing, model-selection, and performance-metric workflows and documenting results.
- Supported production deployment by collaborating with developers to integrate ML models into internal applications.

Indian Institute of Technology, Jodhpur

May 2025 – Jul. 2025

Research Intern – Quantum Computing & Cryptography [GitHub]

Jodhpur, India

- Built a 4-layer QKD security framework in Qiskit simulating BB84 under depolarizing noise, intercept-resend, and PNS attacks, generating 10,000 labeled sessions across 5 attack/channel classes.
- Achieved 80.6% multiclass accuracy and 0.987 binary ROC-AUC using a Random Forest trained on 8 statistical features including QBER, timing jitter, sifting ratio, and QBER variance.
- Built a PyTorch LSTM on 20-step QBER time-series to detect temporal attack patterns and benchmarked it against the Random Forest baseline.
- Designed a closed-loop adaptive defense policy with CONTINUE, HARDEN, and ABORT actions evaluated at up to 4 dynamic checkpoints using predicted attack probabilities.

TECHNICAL SKILLS

Languages: Python, C++, SQL, JavaScript **Quant Finance:** Alpha Research, Statistical Backtesting, Time-Series Analysis, Market Microstructure, Transaction-Cost Modeling, Walk-Forward Validation, Probability & Statistics

Quant Methods: Statistical Modeling, Regression, Classification, Parameter Sensitivity, Out-of-Sample Testing, Risk Analysis **ML/Data:** PyTorch, TensorFlow, scikit-learn, Pandas, NumPy, SciPy

Tools: Git, Linux/macOS, Docker, FastAPI, Next.js, Qiskit, Jupyter

EDUCATION

The LNM Institute of Information Technology

Jaipur, India

B.Tech–M.Tech in Electronics and Communication Engineering

2024 – 2029

- Coursework:** Probability & Statistics, Machine Learning, Linear Algebra, Data Structures & Algorithms, Optimization, Numerical Methods, Stochastic Calculus

ACHIEVEMENTS

- WorldQuant Brain – Gold Level:** Conducted systematic alpha research by developing, backtesting, and refining predictive signals using financial data fields, mathematical operators, cross-sectional ranking, neutralization, and decay. ↗
- Codeforces Expert (rating:1603):** Competitive programming with strong grounding in algorithms, data structures, and problem-solving under time constraints. ↗
- Best Alpha: 1.72 Sharpe, 1.77 Fitness, 13.31% returns, 47.11 bps margin, with 5.65% turnover and 14.79% drawdown;** maintained positive Sharpe across 2019–2023.

CERTIFICATIONS

- Financial Engineering and Risk Management & Term-Structure and Credit Derivatives** — Columbia University
- Citi – Markets Quantitative Analysis (MQA) Job Simulation**